

Biswarup Saha

Stat-Math Unit, Indian Statistical Institute
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EDUCATION

Indian Statistical Institute <i>Senior Research Fellow in Mathematics</i> Advisor: Prof. B. V. Rajarama Bhat	Bangalore, India <i>Aug 2024 – Present</i>
Indian Statistical Institute <i>Junior Research Fellow in Mathematics</i> Coursework Grade: A (Maximum)	Bangalore, India <i>Aug 2022 – Jul 2024</i>
Indian Statistical Institute <i>Master of Mathematics (M.Math)</i> Score: 1764 / 2000	Kolkata, India <i>Dec 2020 – Jun 2022</i>
Presidency University <i>Bachelor of Science in Mathematics (Hons.)</i> Score: 962 / 1000	Kolkata, India <i>Jul 2017 – Jul 2020</i>
Hindu School <i>HSC (10+2 standard)</i> Score: 453 / 500	Kolkata, India <i>May 2017</i>
Narain Dass Bangur Memorial Multipurpose School <i>SSC (10th standard)</i> Score: 633 / 700	Kolkata, India <i>May 2015</i>

RESEARCH INTERESTS

Quantum Information Theory, Operator Algebras, Quantum Graphs.

RESEARCH ARTICLES

- **A complete characterization of eventually entanglement breaking unital quantum channels**
(with B. V. Rajarama Bhat and Pankaj Dey)
arXiv preprint. [arXiv:2609.24168](https://arxiv.org/abs/2609.24168)
- **Spectral radii for subsets of Hilbert C^* -modules and spectral properties of positive maps**
(with B. V. Rajarama Bhat and Prajakta Sahasrabuddhe)
Positivity **30**, 49 (2026). DOI: [10.1007/s11117-026-01207-6](https://doi.org/10.1007/s11117-026-01207-6)

FELLOWSHIPS AND ACHIEVEMENTS

Fellowships & National Qualifications

- 2022** Selected for **NBHM JRF Fellowship** (National Board for Higher Mathematics), DAE, Govt. of India. (Ref: 0203/22(21)/2022/R&D-II/11196 dated on 29.08.2022.)
- 2022** Selected for **Junior Research Fellowship** programme at Indian Statistical Institute.
- 2022** Qualified **CSIR-UGC NET JRF** (Ref: 211610089709 dated on 24.03.2022).
- 2022** Qualified **GATE** (Mathematics).
- 2020** Qualified **NBHM Doctoral Scholarship Test**.
- 2018** Recipient of **INSPIRE Scholarship for Higher Education (SHE)**, DST, Govt. of India.

Academic Excellence (Presidency University)

- 2021 Binapani Mukherjee Memorial Gold Medal** for securing 1st position in B.Sc. Mathematics (2020).
- 2021 Arabinda Mukhopadhyay Memorial Award** for securing 1st position among all Physical Sciences (2020).
- 2021** Recipient of multiple memorial awards for securing 1st position in B.Sc. Mathematics:
The Late Ramanuja Pattu Aiyenger Memorial Award, Shovana Devi & Upendra Narayan Bhattacharya Annual Award, Panchanan Pal Memorial Prize, Late Sushil Kumar Banerji Memorial Award, Dr. Meghnad Saha Memorial Prize, and Bibhuti Bhusan Sen Memorial Award.
- 2020 Dr. Santinath Bagchi Memorial Prize** for 1st position in B.Sc. Mathematics (Years 1 & 2).
- 2019 Bholanath Dass Memorial Award** for 1st position in B.Sc. (Year 1) among all departments.

Selective Admissions & Competitions

- 2020** Qualified **JAM** (Mathematics) with **AIR 42**.
- 2020** Qualified **GS-2020** conducted by TIFR, Mumbai.
- 2020** Selected for Graduate/PhD programs at: **ISI Kolkata** (M.Math), **IIT Kanpur** (M.Sc.), **IMSc Chennai** (Int. PhD), **IISER Bhopal** (Int. PhD), and **IISER Mohali** (Int. PhD).
- 2019 2nd Prize**, Inter-College Poster Presentation Competition, Ramakrishna Mission Vidya-mandira, Belur.
Theme: "Geometry and its Applications" (Feb 18, 2019).

School Level Achievements

- 2017 Laketown Thana Utsav Samannya Committee Award**
Issued by Laketown Police Station (Govt. of West Bengal) for securing 3rd position in Higher Secondary Examination under Laketown Police Station area.

PRESENTATIONS AND SEMINARS

- **Spectral radii for subsets of Hilbert C^* -modules and spectral properties of positive maps**
ISI, Bangalore May 2024
- **Transversality Homotopy Theorem**
ISI, Kolkata Jan 2022
- **Riemann Theorem on Rearrangement of Series**
Presidency University, Kolkata Feb 2019
- **Construction of a Doughnut-shaped Structure Embedded in $\mathbb{R}^3$**
Ramakrishna Mission Vidyamandira, Belur Jan 2018

CONFERENCES AND WORKSHOPS

- **Workshop on Non-commutative Analysis and Probability** Jan 2026
IISc Bangalore (Organisers: Prof. T. Bhattacharyya, Prof. J. M. Lindsay)
- **Quantum Probability Day at ISI** Jan 2025
ISI Bangalore (Organiser: Prof. B. V. R. Bhat)
- **Quantum Analysis I** Jan 2025
IIT Madras (Organisers: Prof. K. K. Mukherjee, Dr. I. Patri, Dr. P. Fima)
- **Noncommutative Mathematics and Applications** Oct 2024
ISI Bangalore (Organisers: Prof. J. Sarkar, Prof. D. Goswami, Prof. A. Pal)
- **International Conference on Spectral and Approximation Theory** Nov 2023
Cochin University of Science And Technology
- **AIS: Function Spaces, Operator Theory and Operator Algebra** Jun 2023
BITS Pilani Goa Campus (Organiser: Prof. J. Sarkar, Dr. M. Bhattacharjee, Dr. S. De)
- **Perspectives in Quantum Probability** May 2023

- ISI Bangalore (Organiser: Prof. B. V. R. Bhat)
- **MTTS Programme (Level I)** May 2020
Funded by NBHM (Online)
 - **Madhava Nurture Camp** Jun 2019
St. Xavier's College, Ahmedabad (Funded by NBHM)

TEACHING EXPERIENCE

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|---|---|
| Teaching Assistant | Indian Statistical Institute, Bangalore |
| • Geometry (B.Math III)
<i>Instructor: Rumpa Masanta</i> | Jan 2026 – May 2026 |
| • Advanced Linear Algebra (M.Math II, B.Math III)
<i>Instructor: B. V. Rajarama Bhat</i> | Jan 2025 – May 2025 |
| • Game, Graph and Algebra (M.Math II, B.Math III)
<i>Instructor: Yogeshwaran Dhandapani</i> | Aug 2024 – Nov 2024 |
| • Analysis on Graphs (B.Math III)
<i>Instructor: B. V. Rajarama Bhat</i> | Aug 2023 – Nov 2023 |

LANGUAGES

Bengali (Native) · **English** (Full Professional Proficiency) · **Hindi** (Basic)

REFERENCES

Prof. B. V. Rajarama Bhat
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 Indian Statistical Institute, Bangalore
 R. V. College Post, Bangalore 560059
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